

BlackRock industry project • UC Berkeley Haas MFE

Team Results: Macro Surprises and DTW on Equal Terms

Four workstreams, four futures markets (ES, NQ, YM, ZN), one set of data, entry rules and costs

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Supervisor: Benjamin Steel, BlackRock

Sources: notebooks 08 and 08b (sections 9–10) • combined/surprise_dtw_book • combined/team_book



What this deck covers



Team comparison at equal costs

slides 4–6

Notebook 08, sections 9–10: each person's own book re-priced on one window (2019–2026) and one Sharpe method, at three cost levels



Surprise and surprise + DTW books

slides 7–8

combined/surprise_dtw_book: the macro-surprise strategies run on the combined DTW book's data, years, costs and metrics



Team book

slides 9–12

combined/team_book: all four workstreams on the same terms, on every market where they exist, combined into team portfolios

Slide 13: conclusions

Four workstreams, one set of rules

Strategy	Built by	Direction from	Markets	Trades / yr (ES)
Macro-surprise book	Reza	consensus surprise, top-10 releases + CPI	ES · NQ · YM · ZN	~96
CPI surprise model	Reza	predicted CPI move, if above cost	ES · NQ · YM · ZN	~11
DTW ladder book	Jack	DTW price-path analogues, 5 windows	ES · NQ · YM · ZN	~310
DTW event-game book	Hoshea	DTW factor on Claims, JOLTS, PCE	ES	~370
Machine-learning classifier	Aaryen	walk-forward classifier, 30 min	ES	~54

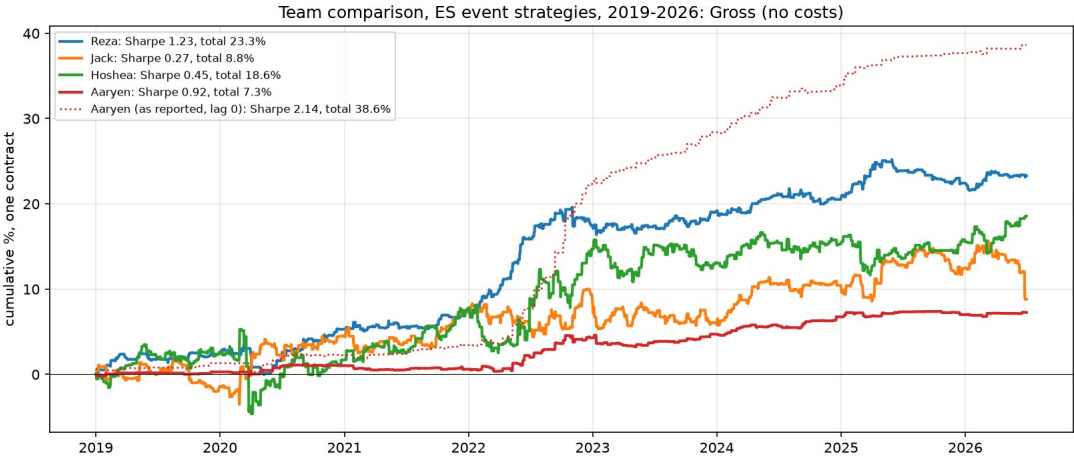
Each person's own positions are kept; only returns, costs and scoring are recomputed. Hoshea's and Aaryen's work exists on ES only. YM (E-mini Dow) was added after the ES and NQ results were known.

The same terms for everyone

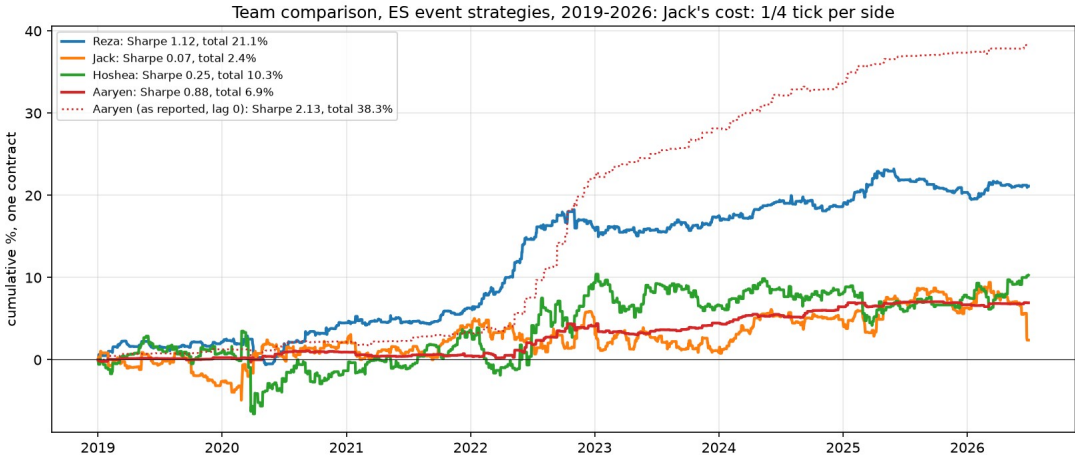
- Jack's cleaned 1-minute bars, roll jumps removed
- Entry only after the release is public (T+1)
- Costs: gross, ¼ tick, ½ tick per side, and 2 ticks + \$4.50
- Walk-forward; validation 2018–20, test 2021–26
- Metrics: daily bps, Sharpe over 252 days, max drawdown

A • Four workstreams at three cost levels

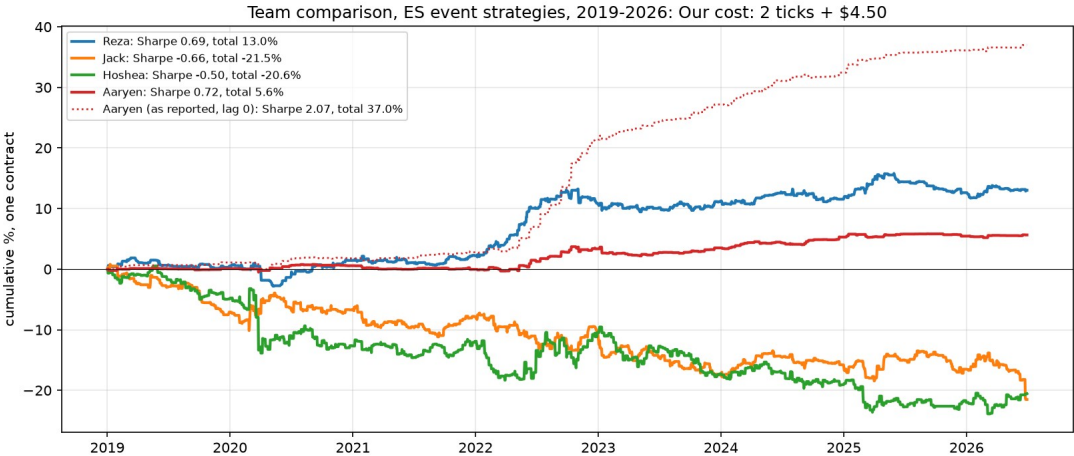
Gross (no costs)



¼ tick per side (Jack's cost)



2 ticks + \$4.50 (our cost)

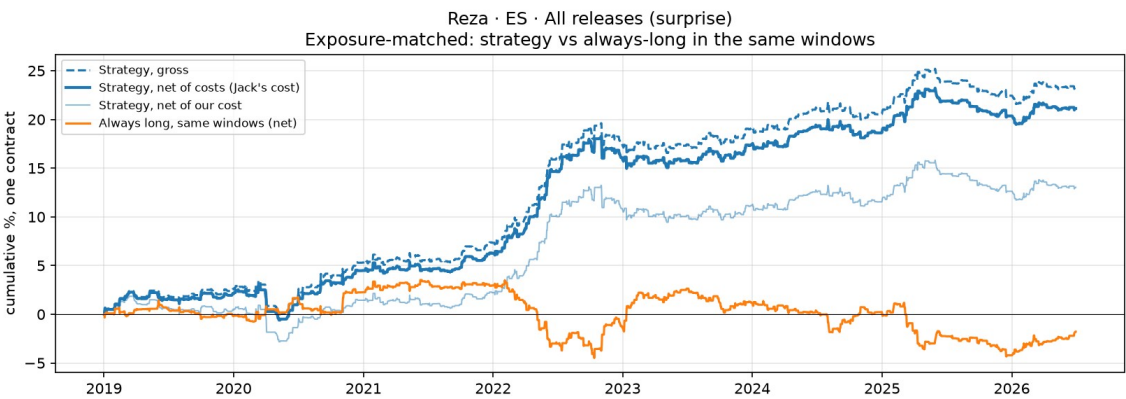


Sharpe / total	Gross	¼ tick	2 ticks + \$4.50
Reza (ES, all releases)	1.23 / 23.3%	1.12 / 21.1%	0.69 / 13.0%
Jack (rebuild)	0.27 / 8.8%	0.07 / 2.4%	-0.66 / -21.5%
Hoshea	0.45 / 18.6%	0.25 / 10.3%	-0.50 / -20.6%
Aaryen (executable)	0.92 / 7.3%	0.88 / 6.9%	0.72 / 5.6%
Aaryen (as reported)	2.14 / 38.6%	2.13 / 38.3%	2.07 / 37.0%

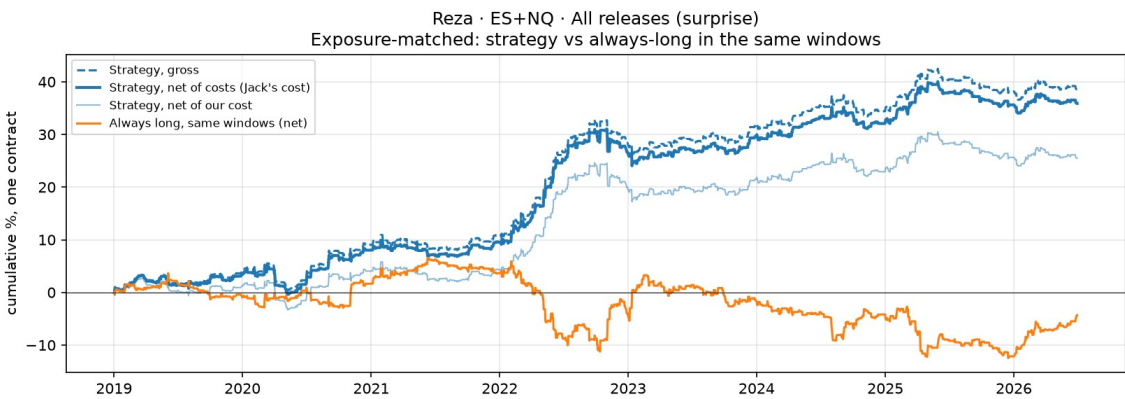
Only the surprise and classifier books keep a positive return at our cost.

A • Each workstream in the peer slide format (1/2)

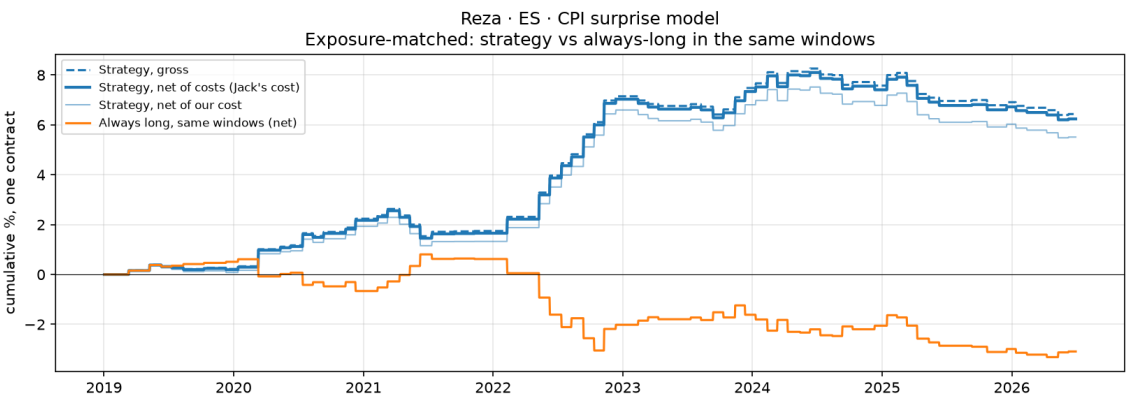
Reza • ES • all releases (surprise)



Reza • ES + NQ • all releases (surprise)



Reza • ES • CPI surprise model

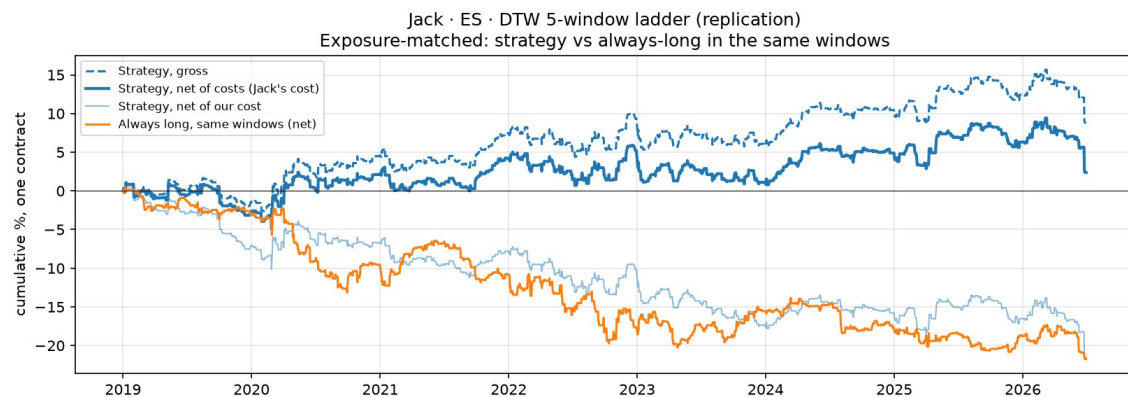


¼ tick per side	ES all	ES + NQ	ES CPI
Total %	21.1	35.9	6.2
Sharpe	1.12	1.06	0.84
Max drawdown	-3.7%	-7.0%	-1.9%
Trades / yr	98	195	9
Always long, same windows	-1.8%	-4.3%	-3.1%

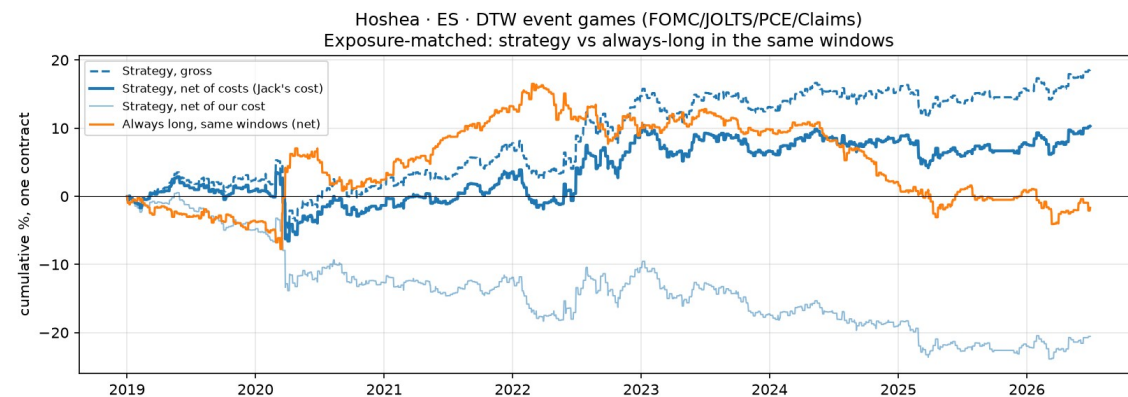
Solid: net; dashed: gross; orange: always long in the same windows.

A • Each workstream in the peer slide format (2/2)

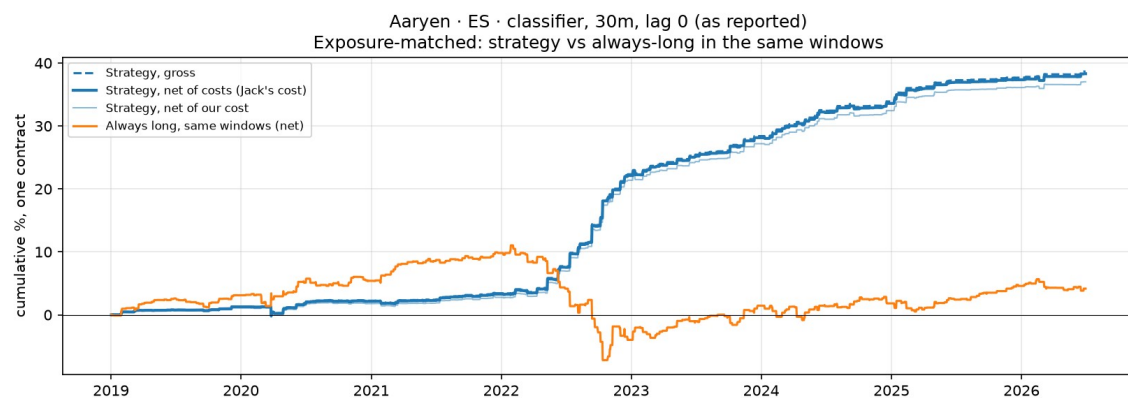
Jack • DTW ladder (rebuild): +2.4%, Sharpe 0.07, 315 trades/yr



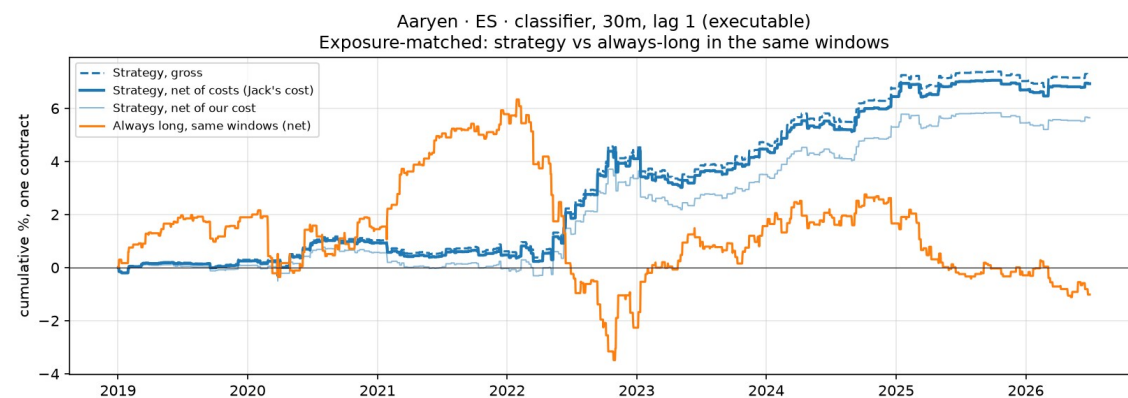
Hoshea • DTW event games: +10.3%, Sharpe 0.25, 372 trades/yr



Aaryen • classifier, as reported: +38.3%, Sharpe 2.13 (not tradeable)



Aaryen • classifier, executable: +6.9%, Sharpe 0.88, 55 trades/yr



B · Surprise books on the combined book's terms

combined/surprise_dtw_book · Sharpe at ¼ tick per side · test years 2021–2026

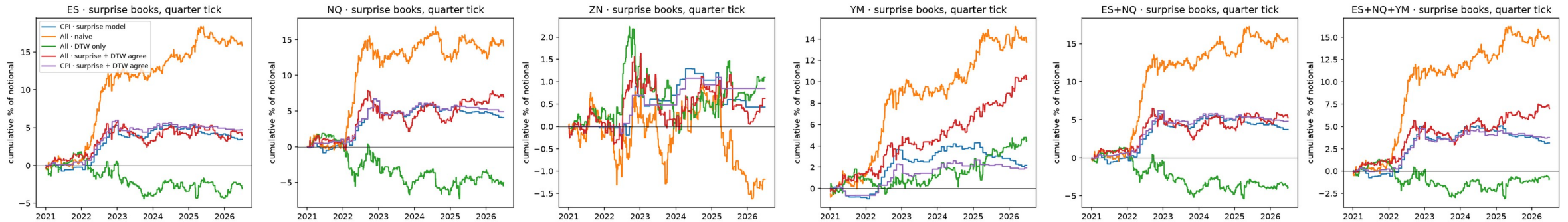
Same terms as the DTW book

- Jack's cleaned bars, costs, session grid, metrics and equal-risk combined book
- Walk-forward from 2013; validation 2018–20, test 2021–26
- One 30-min window at T+1 (the DTW book uses five)
- Five books: two surprise-only, one DTW-only, two surprise + DTW

Book	ES	NQ	ZN	ES+NQ+ZN	ES+NQ	YM	ES+NQ+YM
DTW ladder book (Jack, current)	0.64	0.11	-0.97	0.40	0.54	-0.39	0.27
Surprise, top-10 releases	1.09	0.75	-0.16	0.84	1.00	1.06	1.08
CPI surprise model	0.51	0.46	0.17	0.47	0.49	0.39	0.47
Surprise + DTW agree	0.37	0.48	0.12	0.45	0.48	1.04	0.74
CPI + DTW agree	0.74	0.58	0.48	0.72	0.68	0.41	0.63
DTW only (one window)	-0.26	-0.28	0.19	-0.23	-0.32	0.44	-0.09

- The surprise direction beats the DTW book on all three equity indices with about a third of the trades; ZN fails at any real cost
- DTW row: Jack's current book (commit ac384c3); YM run through the same code (DTW_EXTRA=YM), where it loses (-0.39)

B · Adding DTW and its effect on the results



Sharpe, ¼ tick	ES 2018-20	ES 2021-26	NQ 2018-20	NQ 2021-26	YM 2018-20	YM 2021-26
Surprise, top-10 releases	1.04	1.09	0.86	0.75	1.27	1.06
DTW only	0.57	-0.26	0.78	-0.28	0.40	0.44
Surprise + DTW agree	1.46	0.37	1.21	0.48	0.63	1.04

What this means

- The surprise book is positive in both periods
- In ES and NQ, DTW's direction worked in 2018-20 and reversed after 2021, the same flip Jack found in dtw_dir
- So the DTW filter looked best in validation and worst in test; on YM it neither helps nor hurts in test (1.04 vs 1.06)
- Most of the surprise book's gain came in 2022

C · Team book: coverage and overlap

Strategy	ES	NQ	YM	ZN
Macro-surprise book	✓	✓	✓	✓
CPI surprise model	✓	✓	✓	✓
DTW ladder book	✓	✓	✓	✓
DTW event-game book	✓	–	–	–
Machine-learning classifier	✓	–	–	–

Hoshea's and Aaryen's work exists on ES only; the team portfolios use every book on every market where it exists.

Correlation of daily P&L, 2021–26	value
Surprise ES vs surprise NQ (same releases)	0.68
Surprise ES vs surprise YM (same releases)	0.85
DTW ladder ES vs DTW ladder NQ	0.15
Classifier vs surprise ES	0.27
Classifier vs surprise ZN	0.31
DTW event games vs every other book	–0.04 to 0.05
Surprise vs DTW ladder (any market)	–0.07 to 0.16

Across people the books barely move together: they trade different releases and windows, which is what makes a team portfolio worth building.

C · Each strategy on each market

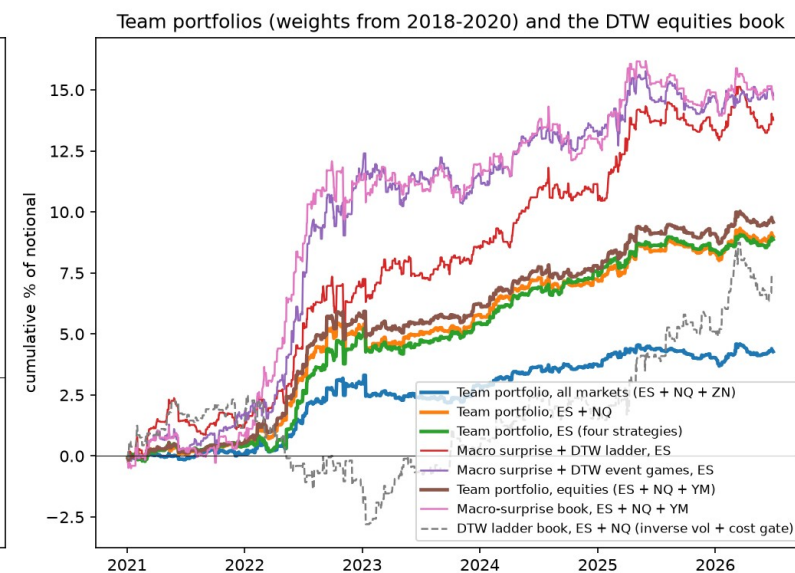
Sharpe, test years 2021–2026 · “–” = strategy not available on that market

Strategy	ES ¼ tick	NQ ¼ tick	ZN ¼ tick	YM ¼ tick	ES 2t+\$4.5	NQ 2t+\$4.5	ZN 2t+\$4.5	YM 2t+\$4.5	ES 2018–20
Macro-surprise book	1.09	0.75	-0.16	1.06	0.74	0.64	-1.51	0.82	1.04
CPI surprise model	0.51	0.46	0.17	0.39	0.61	0.45	-0.43	0.29	1.27
DTW ladder book (Jack)	0.64	0.11	-0.97	-0.39	0.10	-0.04	-5.03	-0.74	0.59
DTW event-game book	0.42	–	–	–	-0.29	–	–	–	0.26
Machine-learning classifier	0.95	–	–	–	0.81	–	–	–	0.38
Buy and hold (no costs)	0.69	0.58	-0.69	0.67	–	–	–	–	0.80

- At ¼ tick the macro-surprise book leads on ES, NQ and YM; at a realistic cost the surprise book and the classifier hold up, the high-turnover DTW books do not
- ZN loses after costs in every strategy; the DTW ladder loses on YM even before costs (run with Jack's exact code)

C • The strategies and the team portfolios over time

Macro-release strategies on ES, NQ, ZN, YM, 2021-2026, a quarter tick per side



ES

- Surprise book climbs most, largely in 2022
- Classifier: small but steady

NQ, ZN and YM

- Surprise NQ and YM climb with ES
- ZN books drift down: the cost on every trade

Team portfolios

- ES, ES + NQ and ES + NQ + YM: steady, small drawdowns
- Adding ZN (blue) lowers the whole book

C · Team portfolios

Equal risk, weights from 2018–2020 only · test years 2021–2026 · return, vol and drawdown at ¼ tick

Portfolio	Gross	¼ tick	½ tick	2t + \$4.50	Return	Vol	Max DD	Trades/yr
Team, equities (ES + NQ + YM)	1.48	1.35	1.22	0.85	+9.6%	1.3%	–1.1%	1,131
Team, ES + NQ	1.45	1.31	1.17	0.78	+9.0%	1.3%	–1.1%	1,043
Team, ES (four strategies)	1.55	1.39	1.22	0.76	+8.9%	1.2%	–1.0%	739
Team, all markets	1.35	0.83	0.30	–1.09	+4.3%	1.0%	–1.1%	1,562
Surprise + DTW event games (ES)	1.32	1.14	0.96	0.46	+14.8%	2.4%	–2.2%	465
Surprise + DTW ladder (ES)	1.31	1.16	1.01	0.59	+13.8%	2.2%	–2.0%	315
Macro-surprise book alone (ES)	1.18	1.09	0.99	0.74	+15.9%	2.7%	–2.6%	96
Macro-surprise book (ES + NQ + YM)	1.13	1.07	1.01	0.82	+14.6%	2.6%	–2.4%	280
ES buy and hold	0.69	–	–	–	+62.4%	16.9%	–31.1%	–

At ¼ tick, combining helps

The ES team book reaches 1.39 and the equity book (ES + NQ + YM) 1.35, above the best single book (1.09) with half its volatility and drawdowns near –1%.

At a realistic cost, the equity book holds up best

At 2 ticks + \$4.50 the ES + NQ + YM team book keeps 0.85, ahead of the surprise book alone (0.74 on ES); the DTW books trade 300+ times a year, and adding ZN turns the book negative.

Conclusions

1

The macro-surprise direction is the strongest single signal in equities

ES 1.09, NQ 0.75 and YM 1.06 at $\frac{1}{4}$ tick; positive in both validation and test years

2

DTW works for sizing, not direction

Its direction flips between 2018–20 and 2021–26 in ES and NQ, and on YM the DTW book loses (−0.39)

3

The four workstreams are almost uncorrelated

Team portfolios reach 1.39 (ES) and 1.35 (ES + NQ + YM) at $\frac{1}{4}$ tick with half the risk

4

Execution cost decides the result

At 2 ticks + \$4.50 the surprise book, the classifier and the equity team book (0.85) stay positive; ZN fails everywhere

Caveats: Hoshea's setting was chosen on the full sample; the test years have been examined several times; much of the gain comes from 2022; YM was added after the ES and NQ results were known.

Thank you

Questions and discussion

Team code and results: github.com/jack-duncan/blackrock-intraday · Reza's study and papers: github.com/rezazamani2329/BlackRock-Project-State-Dependent-Intraday-Response-of-S-P-500-Futures-to-Macroeconomic-Surprises